

SCREEN SHOTS:

SECRETEXEC

```
• (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py compile examples/demo/arrays.c
Successfully compiled 'examples/demo/arrays.c' -> 'examples/demo/arrays.bc'
Generated 39 instructions.
```

```
• (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py run examples/demo/arrays.bc
Verification:
✓ arrays.bc SAFE

Execution completed.

-----
Process Report

PID:
1

Program:
arrays.bc

Status:
TERMINATED

Exit Code:
0

Instructions:
35

CPU Time:
35

Memory:

Virtual Memory:
256 cells

Heap:
0 cells

Stack:
0 cells

Program Output:

6

-----

Execution Summary:

Programs Submitted:
1

Verification Passed:
1

Verification Failed:
0
```

```
Scheduler Statistics:

Processes Created:
1

Processes Completed:
1

Context Switches:
1

Average Waiting Time:
0.0

Average Turnaround Time:
35.0

CPU Utilization:
100.0%
```

```
(venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py execute examples/demo/hello.c
SecureExec Runtime v1.0

Compilation:
✓ hello.c

Verification:
✓ hello.c SAFE

Execution:

Process:
hello.c

PID:
1

Output:
1337

Execution Summary:

Programs Submitted:
1

Verification Passed:
1

Verification Failed:
0

Scheduler Statistics:

Processes Created:
1

Processes Completed:
1

Context Switches:
1

Average Waiting Time:
0.0

Average Turnaround Time:
10.0

CPU Utilization:
100.0%
```

```
(venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py execute examples/demo/hello.c --debug
SecureExec Runtime v1.0

Compilation:
[Compiler]: Compiling examples/demo/hello.c
✓ hello.c

Verification:
✓ hello.c SAFE

Execution:

PID 1 entered Q1
[Scheduler]: Starting execution of 1 processes
From Q1:
PID 1 Used: 5 instructions
Moving PID 1 to Q2
From Q2:
PID 1 Used: 5 instructions
Process PID 1 completed and removed
Process:
hello.c

PID:
1

Output:
1337

Registers:
R0: 0
R1: 0
R2: 0
R3: 0
R4: 0
R5: 0
R6: 0
R7: 0
R8: 0
R9: 0
R10: 0
R11: 0
R12: 0
R13: 0
R14: 0
R15: 0

Execution Summary:

Programs Submitted:
1

Verification Passed:
1
```

```
Verification Passed:
1

Verification Failed:
0

Scheduler Statistics:

Processes Created:
1

Processes Completed:
1

Context Switches:
1

Average Waiting Time:
0.0

Average Turnaround Time:
10.0

CPU Utilization:
100.0%
```

```
(venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py execute examples/demo/hello.c examples/demo/arrays.c examples/demo/heap.c
SecureExec Runtime v1.0

Compilation:
✓ hello.c
✓ arrays.c
✓ heap.c

Verification:
✓ hello.c SAFE
✓ arrays.c SAFE
✓ heap.c SAFE

Execution:

Process:
hello.c

PID:
1

Output:
1337

Process:
arrays.c

PID:
2

Output:
6

Process:
heap.c

PID:
3

Output:
30

Execution Summary:

Programs Submitted:
3

Verification Passed:
3

Verification Failed:
0
```

```
Verification Failed:
0

Scheduler Statistics:

Processes Created:
3

Processes Completed:
3

Context Switches:
16

Average Waiting Time:
31.67

Average Turnaround Time:
61.0

CPU Utilization:
100.0%
```

```
(venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py execute examples/demo/unsafe_but_valid.c
SecureExec Runtime v1.0

Compilation:
✓ unsafe_but_valid.c

Verification:
X unsafe_but_valid.c REJECTED

Rejected Programs:

Program:
unsafe_but_valid.c

Reason:
-----
SecureExec Verification Report
Program: unsafe_but_valid.c
Status: REJECTED

Checks:
X Instruction validity

Security Score: 60/100

Resource Risk:
HIGH

Loop Analysis:
Detected loops:
1

Bounded loops:
0

Estimated maximum iterations:
0

Reason:
Potential infinite execution detected.

Details:
Loop condition variable:
1

Initial value:
0

Modification inside loop:
None

Warnings:
⚠ Instruction 99 is unreachable.
```

```
Warnings:
⚠ Instruction 99 is unreachable.
⚠ Instruction 100 is unreachable.
⚠ Instruction 101 is unreachable.
⚠ Instruction 102 is unreachable.
-----
```

```
(venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py execute examples/demo/hello.c examples/demo/arrays.c examples/demo/heap.c examples/demo/unsafe_but_valid.c
SecureExec Runtime v1.0

Compilation:
✓ hello.c
✓ arrays.c
✓ heap.c
✓ unsafe_but_valid.c

Verification:
✓ hello.c SAFE
✓ arrays.c SAFE
✓ heap.c SAFE
X unsafe_but_valid.c REJECTED

Rejected Programs:

Program:
unsafe_but_valid.c

Reason:
-----
SecureExec Verification Report
Program: unsafe_but_valid.c
Status: REJECTED

Checks:
X Instruction validity

Security Score: 60/100

Resource Risk:
HIGH

Loop Analysis:
Detected loops:
1

Bounded loops:
0

Estimated maximum iterations:
0

Reason:
Potential infinite execution detected.

Details:
Loop condition variable:
1

Initial value:
0
```

```
Modification inside loop:
None

Warnings:
⚠Instruction 99 is unreachable.
⚠Instruction 100 is unreachable.
⚠Instruction 101 is unreachable.
⚠Instruction 102 is unreachable.
```

Execution:

Process:
hello.c

PID:
1

Output:
1337

Process:
arrays.c

PID:
2

Output:
6

Process:
heap.c

PID:
3

Output:
30

Execution Summary:

Programs Submitted:
4

Verification Passed:
3

Verification Failed:
1

Scheduler Statistics:

Processes Created:

Scheduler Statistics:

Processes Created:
3

Processes Completed:
3

Context Switches:
16

Average Waiting Time:
31.67

Average Turnaround Time:
61.0

CPU Utilization:
100.0%

```
• (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py compile examples/demo/functions.c
Successfully compiled 'examples/demo/functions.c' -> 'examples/demo/functions.bc'
Generated 40 instructions.
• (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py run examples/demo/heap.bc examples/demo/functions.bc examples/demo/arrays.bc
Verification:
✓ heap.bc SAFE
✓ functions.bc SAFE
✓ arrays.bc SAFE

Execution completed.

-----
Process Report

PID:
1

Program:
heap.bc

Status:
TERMINATED

Exit Code:
0

Instructions:
43

CPU Time:
43

Memory:

Virtual Memory:
272 cells

Heap:
1 cells

Stack:
0 cells

Program Output:

30

-----
```

Process Report

PID:
2

Program:
functions.bc

Status:
TERMINATED

Exit Code:
0

Instructions:
32

CPU Time:
32

Memory:

Virtual Memory:
256 cells

Heap:
0 cells

Stack:
0 cells

Program Output:

30

Process Report

PID:
3

Program:
arrays.bc

Status:
TERMINATED

Exit Code:
0

Instructions:

```
Instructions:
35

CPU Time:
35

Memory:

Virtual Memory:
256 cells

Heap:
0 cells

Stack:
0 cells

Program Output:

6

-----

Execution Summary:

Programs Submitted:
3

Verification Passed:
3

Verification Failed:
0

Scheduler Statistics:

Processes Created:
3

Processes Completed:
3

Context Switches:
22

Average Waiting Time:
66.33

Average Turnaround Time:
103.0

CPU Utilization:
100.0%
```

```
• (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py execute examples/demo/hello.c examples/demo/arrays.c examples/demo/heap.c examples/demo/unsafe_but_valid.c --debug
SecureExec Runtime v1.0

Compilation:
[Compiler]: Compiling examples/demo/hello.c
✓ hello.c
[Compiler]: Compiling examples/demo/arrays.c
✓ arrays.c
[Compiler]: Compiling examples/demo/heap.c
✓ heap.c
[Compiler]: Compiling examples/demo/unsafe_but_valid.c
✓ unsafe_but_valid.c

Verification:
✓ hello.c SAFE
✓ arrays.c SAFE
✓ heap.c SAFE
X unsafe_but_valid.c REJECTED

Rejected Programs:

Program:
unsafe_but_valid.c

Reason:
-----
SecureExec Verification Report
Program: unsafe_but_valid.c
Status: REJECTED

Checks:
X Instruction validity

Security Score: 60/100

Resource Risk:
HIGH

Loop Analysis:
Detected loops:
1

Bounded loops:
0

Estimated maximum iterations:
0

Reason:
Potential infinite execution detected.

Details:
Loop condition variable:
```

```
Details:
Loop condition variable:
1

Initial value:
0

Modification inside loop:
None

Warnings:
⚠ Instruction 99 is unreachable.
⚠ Instruction 100 is unreachable.
⚠ Instruction 101 is unreachable.
⚠ Instruction 102 is unreachable.

-----

Execution:

PID 1 entered Q1
PID 2 entered Q1
PID 3 entered Q1
[Scheduler]: Starting execution of 3 processes
From Q1:
PID 1 Used: 5 instructions
Moving PID 1 to Q2
Context Switch:
PID 1 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Moving PID 2 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Context Switch:
PID 3 -> PID 1
From Q1:
PID 1 Used: 5 instructions
Process PID 1 completed and removed
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 1 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Moving PID 2 to Q2
Context Switch:
PID 2 -> PID 3
From Q1:
```

```
Moving PID 2 to Q2
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 3 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Moving PID 2 to Q2
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 3 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Moving PID 2 to Q2
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 3 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Moving PID 2 to Q2
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 3 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Moving PID 2 to Q2
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
```

```
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
Context Switch:
PID 3 -> PID 2
From Q1:
PID 2 Used: 5 instructions
Process PID 2 completed and removed
Context Switch:
PID 2 -> PID 3
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
Starvation prevention:
Moving Q2 processes back to Q1
From Q1:
PID 3 Used: 5 instructions
Moving PID 3 to Q2
From Q2:
PID 3 Used: 3 instructions
Process PID 3 completed and removed
Process:
hello.c

PID:
1

Output:
1337

Registers:
R0: 0
R1: 0
R2: 0
R3: 0
R4: 0
R5: 0
R6: 0
R7: 0
R8: 0
R9: 0
R10: 0
R11: 0
R12: 0
R13: 0
R14: 0
R15: 0

Process:
arrays.c

PID:
2
```

```
PID:
2

Output:
6

Registers:
R0: 0
R1: 3
R2: 3
R3: 0
R4: 0
R5: 0
R6: 0
R7: 0
R8: 0
R9: 0
R10: 0
R11: 0
R12: 0
R13: 0
R14: 0
R15: 0

Process:
heap.c

PID:
3

Output:
30

Registers:
R0: 0
R1: 10
R2: 20
R3: 0
R4: 0
R5: 0
R6: 0
R7: 0
R8: 0
R9: 0
R10: 0
R11: 0
R12: 0
R13: 0
R14: 0
R15: 0

Execution Summary:
Programs Submitted:
```

```
Execution Summary:

Programs Submitted:
4

Verification Passed:
3

Verification Failed:
1

Scheduler Statistics:

Processes Created:
3

Processes Completed:
3

Context Switches:
16

Average Waiting Time:
31.67

Average Turnaround Time:
61.0

CPU Utilization:
100.0%

> (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51>
```

```
• (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> python cli.py compile examples/demo/functions.c examples/demo/arrays.c  
SecureExec Compiler v1.0
```

```
Compilation:
```

```
✓ functions.c  
Generated:  
functions.bc
```

```
✓ arrays.c  
Generated:  
arrays.bc
```

```
Compilation Summary:
```

```
Programs Submitted:  
2
```

```
Successful:  
2
```

```
Failed:  
0
```

```
❖ (venv) PS C:\Users\jasmi\Downloads\OS PROJECT 51> 
```